MAE-263F Sample Midterm 2

Fall 2026

Monday, November 23, 2026

FULL NAME: _____

UID: _____

Time: 100 minutes (12:05 PM - 1:45 PM PST)

Maximum points: 15.5 (31 questions × 0.5 pts)
Note: This sample midterm has 31 problems. The actual midterm will have 20 problems, each worth 0.5 pts, for a maximum of 10 points.

Please read the following instructions carefully:

1. This sample exam has 31 multiple-choice questions (the actual midterm has 20, each worth 0.5 pts). Each question is on its own page. For each question, **circle exactly ONE answer** (a, b, c, or d). **Only your a/b/c/d answer is graded.** Each question has space below it to show your work; **this work is NOT graded**.
2. You are allowed to use a calculator (including scientific and graphing) for mathematical operations, including finding roots of polynomials. Computers and internet access are NOT allowed.
3. This exam is open notes: you may bring ONE letter-sized two-sided cheat sheet and a printout of the course notes. You may write/type anything you wish on the cheat sheet.
4. Once you are done, please use your phone to **scan your midterm and upload it to Gradescope** before leaving the exam hall. Do NOT write anything once the exam is over at 1:45 pm. However, you may scan and upload after 1:45 pm.
5. Please make sure to **hand over the physical copy** of your midterm to the instructors.
6. If you are unable to scan your exam, please inform the instructor/TA and simply hand over the physical copy.

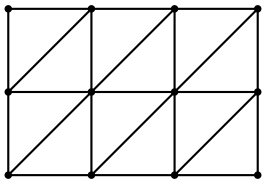
Honor Policy Statement

I agree to the UCLA student code of conduct for academic integrity. I affirm that I will not give or receive unauthorized help on this exam, and all work will be my own. I will not work on the exam outside the allowed time (12:05 pm - 1:45 pm PST).

Signature: _____ Date: _____

Question 1 **0.5 pts**

A rectangular plate with 4×3 nodes is meshed with 12 triangles as shown below (each square is cut by the diagonal from its lower-left to its upper-right corner).



How many edges and how many hinges does the mesh have?

(a) 23 edges and 23 hinges

(b) 17 edges and 7 hinges

(c) 23 edges and 13 hinges

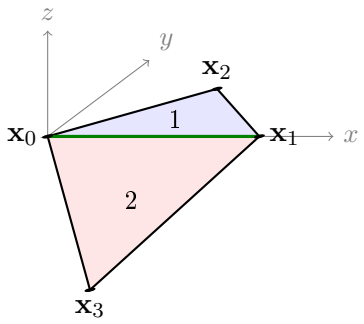
(d) 26 edges and 13 hinges

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 2 **0.5 pts**

Consider the following hinge. The hinge has four nodes (in meters): $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [2, 0, 0]$, $\mathbf{x}_2 = [1, 1, 0]$, and $\mathbf{x}_3 = [1, -1, -1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



Compute the triangle normals $\mathbf{n}_0$ and $\mathbf{n}_1$.

(a) $\mathbf{n}_0 = [0, 0, 2]$, $\mathbf{n}_1 = [0, 2, -2]$

(b) $\mathbf{n}_0 = [0, 0, 2]$, $\mathbf{n}_1 = [0, -2, 2]$

(c) $\mathbf{n}_0 = [0, 0, 1]$, $\mathbf{n}_1 = [0, -1, -1]$

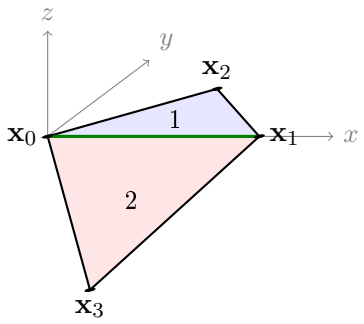
(d) $\mathbf{n}_0 = [2, 0, 0]$, $\mathbf{n}_1 = [0, -2, 2]$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 3 **0.5 pts**

Consider the same hinge as in Question 2 (repeated here). The hinge has four nodes (in meters): $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [2, 0, 0]$, $\mathbf{x}_2 = [1, 1, 0]$, and $\mathbf{x}_3 = [1, -1, -1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



The hinge angle θ is the signed angle from $\mathbf{n}_0$ to $\mathbf{n}_1$ about the hinge vector $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$: it is positive if $\mathbf{n}_0 \times \mathbf{n}_1$ points along $\mathbf{e}^0$ and negative otherwise. What is θ?

(a) $-\pi/4$

(b) $\pi/2$

(c) $3\pi/4$

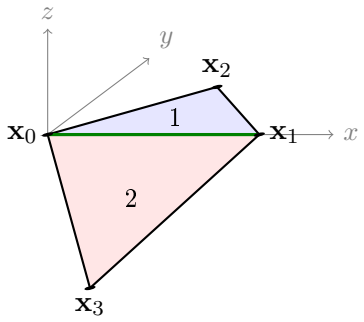
(d) $\pi/4$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 4 **0.5 pts**

Consider the same hinge as in Questions 2–3 (repeated here), whose hinge angle is $\theta = \pi/4$. The hinge has four nodes (in meters): $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [2, 0, 0]$, $\mathbf{x}_2 = [1, 1, 0]$, and $\mathbf{x}_3 = [1, -1, -1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



The bending stiffness is $k_b = 2\,\mathrm{N\,m}$ and the bending energy is $E_b = \frac{1}{2} k_b (\theta - \bar{\theta})^2$. What is E_b if the hinge is a *plate* ($\bar{\theta} = 0$), and if it is a *shell* with $\bar{\theta} = \pi/6$?

(a) Plate: $\pi^2/8\,\mathrm{J}$; shell: $\pi^2/72\,\mathrm{J}$

(b) Plate: $\pi^2/16\,\mathrm{J}$; shell: $25\pi^2/144\,\mathrm{J}$

(c) Plate: $\pi^2/16\,\mathrm{J}$; shell: $\pi^2/144\,\mathrm{J}$

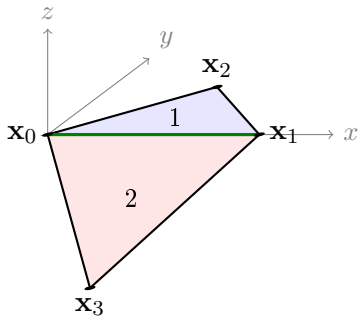
(d) Plate: $\pi/4\,\mathrm{J}$; shell: $\pi/12\,\mathrm{J}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 5 **0.5 pts**

Consider a setup similar to Question 4 (repeated here): a *plate* ($\bar{\theta} = 0$) with $k_b = 2\,\mathrm{N\,m}$ and $E_b = \frac{1}{2}k_b\theta^2$. The hinge has four nodes (in meters): $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, $\mathbf{x}_2 = [1,1,0]$, and $\mathbf{x}_3 = [1,-1,-1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



Node $\mathbf{x}_3$ is now moved from $[1,-1,-1]$ to $[1,0,-1]$, and all other nodes stay where they are. What is the new bending energy E_b?

(a) $\pi^2/4\,\mathrm{J}$

(b) 0

(c) $\pi^2/16\,\mathrm{J}$

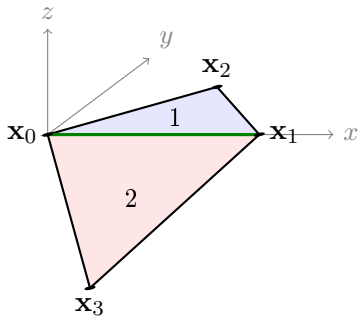
(d) $\pi^2/2\,\mathrm{J}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 6 **0.5 pts**

Consider the same hinge as in Questions 2–4 (repeated here), as a plate with $E_b = \frac{1}{2}k_b\theta^2$. The hinge has four nodes (in meters): $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [2, 0, 0]$, $\mathbf{x}_2 = [1, 1, 0]$, and $\mathbf{x}_3 = [1, -1, -1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



Which component of the gradient of the bending energy is **zero**?

(a) $\partial E_b/\partial y_3$

(b) $\partial E_b/\partial z_0$

(c) $\partial E_b/\partial z_2$

(d) $\partial E_b/\partial y_2$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 7 **0.5 pts**

Let $\mathbf{f}(x,y) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} x^2 + 3y \\ xy^2 \end{bmatrix}$. At $(x,y) = (1,1)$, estimate the Jacobian $J_{ij} = \partial f_i / \partial q_j$ (with $q_1 = x$, $q_2 = y$) using the forward finite difference

$$\frac{\partial f_i}{\partial q_j} \approx \frac{f_i(\mathbf{q} + \Delta q\, \mathbf{e}_j) - f_i(\mathbf{q})}{\Delta q},$$

where $\mathbf{e}_j$ perturbs only q_j, with $\Delta q = 0.1$. That is, perturb x to 1.1 (keeping $y = 1$) for the first column, and perturb y to 1.1 (keeping $x = 1$) for the second column. What is the finite-difference estimate of $\mathbf{J}$?

(a) $\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 2.1 & 1 \\ 3 & 2.1 \end{bmatrix}$

(c) $\begin{bmatrix} 0.21 & 0.3 \\ 0.1 & 0.21 \end{bmatrix}$

(d) $\begin{bmatrix} 2.1 & 3 \\ 1 & 2.1 \end{bmatrix}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 8 **0.5 pts**

An edge of a plate has undeformed length $l = 0.1\,\mathrm{m}$. The plate has Young's modulus $Y = 1\,\mathrm{MPa}$ and thickness $h = 1\,\mathrm{mm}$, so $k_s = \frac{\sqrt{3}}{2} Y h l^2 = 5\sqrt{3}\,\mathrm{N\,m}$. The edge is stretched to a length of $0.11\,\mathrm{m}$. What is its stretching energy $E_s = \frac{1}{2} k_s \left(\frac{\|\mathbf{e}\|}{l} - 1 \right)^2$?

(a) $\sqrt{3}/20\,\mathrm{J}$

(b) $\sqrt{3}/40\,\mathrm{J}$

(c) $\sqrt{3}/4\,\mathrm{J}$

(d) $\sqrt{3}/4000\,\mathrm{J}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 9 **0.5 pts**

The thickness h of a plate is doubled. By what factors do the bending stiffness $k_b = \frac{2}{\sqrt{3}}\frac{Yh^3}{12}$ and the stretching stiffness $k_s = \frac{\sqrt{3}}{2}Yhl^2$ change?

(a) Both double.

(b) Both increase by a factor of 8.

(c) k_b doubles; k_s increases by a factor of 8.

(d) k_b increases by a factor of 8; k_s doubles.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 10 **0.5 pts**

A discrete plate has N nodes. How many DOFs does it have, and what are the sizes of the gradient vector of one hinge's bending energy and of one edge's stretching energy?

(a) $3N$ DOFs; hinge gradient: 12; edge gradient: 6

(b) $4N - 1$ DOFs; hinge gradient: 12; edge gradient: 6

(c) $3N$ DOFs; hinge gradient: 9; edge gradient: 6

(d) $3N$ DOFs; hinge gradient: 12; edge gradient: 3

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 11 **0.5 pts**

In a discrete shell simulation, how is the natural hinge angle $\bar{\theta}$ of each hinge obtained?

(a) It is recomputed at every time step from the current shape.

(b) It is computed once, from the initial (undeformed) shape, before time stepping starts; for a plate, it is zero.

(c) It is zero for every hinge of a shell.

(d) It is a degree of freedom solved for by Newton's method.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 12 **0.5 pts**

A discrete shell has four nodes, $\mathbf{x}_0$, $\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_3$, each with three coordinates. Its DOF vector is $\mathbf{q} = [\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3]$ (size 12, Python indices 0–11), so node $\mathbf{x}_k$ occupies `q[3*k : 3*k+3]`. Nodes $\mathbf{x}_0$, $\mathbf{x}_1$, $\mathbf{x}_2$ are clamped (fixed) and $\mathbf{x}_3$ is free. Which Python indices of $\mathbf{q}$ are the free DOFs?

(a) `0, 1, 2`

(b) `0` through `8`

(c) `9, 10, 11`

(d) All 12 indices

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 13 **0.5 pts**

In resistive force theory, the drag on a rod depends on the direction of motion: moving along the rod is resisted less than moving sideways. So the velocity $\mathbf{v}$ (a column vector) of a node is decomposed into a tangential velocity $\mathbf{v}_t$, along the unit tangent $\mathbf{t}$ (a column vector), and a normal velocity $\mathbf{v}_n$, perpendicular to $\mathbf{t}$, with $\mathbf{v} = \mathbf{v}_t + \mathbf{v}_n$. What is $\mathbf{v}_t$?

(a) $(\mathbf{t}^T\mathbf{t})\,\mathbf{v}$

(b) $(\mathbf{I} - \mathbf{t}\mathbf{t}^T)\,\mathbf{v}$

(c) $\mathbf{t} \times \mathbf{v}$

(d) $(\mathbf{t}\mathbf{t}^T)\,\mathbf{v}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 14 **0.5 pts**

As in Question 13, the velocity $\mathbf{v}$ of a node of a rod is decomposed into a tangential velocity $\mathbf{v}_t$ (along the unit tangent $\mathbf{t}$) and a normal velocity $\mathbf{v}_n$ (perpendicular to $\mathbf{t}$), with $\mathbf{v} = \mathbf{v}_t + \mathbf{v}_n$. For $\mathbf{t} = [0.6, 0.8, 0]^T$ and $\mathbf{v} = [1, 2, 0]^T$ m/s, what is the normal velocity $\mathbf{v}_n$?

(a) $[1.32, 1.76, 0]^T$ m/s

(b) $[-0.32, 0.24, 0]^T$ m/s

(c) $[0.32, -0.24, 0]^T$ m/s

(d) $[0.4, 1.2, 0]^T$ m/s

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 15 **0.5 pts**

In resistive force theory, the drag on node k is $\mathbf{f}_k = -l_k \left(C_t \mathbf{P}_t + C_p \mathbf{P}_n\right) \mathbf{u}_k$, with $\mathbf{P}_t = \mathbf{t}\mathbf{t}^T$ and $\mathbf{P}_n = \mathbf{I} - \mathbf{P}_t$. Let $l_k = 0.01\,\text{m}$, $C_t = 1$, $C_p = 2$ (in $\text{Pa} \cdot \text{s}$), $\mathbf{t} = [1, 0, 0]^T$, and $\mathbf{u}_k = [0.1, 0.1, 0]^T\,\text{m/s}$ (no background flow). What is $\mathbf{f}_k$?

(a) $[-0.001, -0.002, 0]^T\,\text{N}$

(b) $[-0.002, -0.001, 0]^T\,\text{N}$

(c) $[0.001, 0.002, 0]^T\,\text{N}$

(d) $[-0.1, -0.2, 0]^T\,\text{N}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 16 **0.5 pts**

At high Reynolds number, the drag on node k of a rod is $\mathbf{f}_k = -k_k \|\mathbf{u}_k^{\text{rel}}\| \, \mathbf{u}_k^{\text{rel}}$. Here $\mathbf{u}_k^{\text{rel}} = \mathbf{u}_k - \mathbf{U}_\infty$ is the velocity of the node relative to the background flow, and $k_k = \frac{1}{2}\rho C_D A_k$ is a constant that depends on the fluid density ρ, the drag coefficient C_D, and the projected area A_k of the rod segment around node k. If the relative velocity is tripled (same direction), by what factor does the magnitude of the drag force change?

(a) 3

(b) 6

(c) 9

(d) 27

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 17 **0.5 pts**

A rod of diameter $L = 0.01\,\text{m}$ moves at $v = 1\,\text{m/s}$ in water ($\rho = 1000\,\text{kg/m}^3$, $\mu = 10^{-3}\,\text{Pa}\cdot\text{s}$). What is the Reynolds number $Re = \rho v L/\mu$, and which hydrodynamic model is appropriate?

(a) $Re = 10$; resistive force theory (low Re)

(b) $Re = 10^4$; resistive force theory (low Re)

(c) $Re = 10^{-4}$; quadratic drag (high Re)

(d) $Re = 10^4$; quadratic drag (high Re)

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 18 **0.5 pts**

A rod has nodes $\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{N-1}$. The two end nodes, $\mathbf{x}_0$ and $\mathbf{x}_{N-1}$, belong to only one edge each; all other nodes are *interior* nodes and belong to two edges (the edge before and the edge after). The tangent at an interior node $\mathbf{x}_k$ is the normalized average of the unit tangents of these two edges. For an interior node with $\mathbf{x}_k - \mathbf{x}_{k-1} = [2, 0, 0]$ and $\mathbf{x}_{k+1} - \mathbf{x}_k = [0, 3, 0]$, what is the node tangent?

(a) $\frac{1}{\sqrt{2}}[1, 1, 0]$

(b) $\frac{1}{\sqrt{13}}[2, 3, 0]$

(c) $[1, 1, 0]$

(d) $[0, 1, 0]$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 19 **0.5 pts**

In the predictor–corrector method, a node is held fixed on the ground. After the solve, its reaction force points *toward* the ground, i.e., the ground would have to pull the node down to keep it there. What should the algorithm do?

(a) Keep the node fixed, because it is already on the ground.

(b) Release the node (make its vertical DOF free again), because the ground can only push, not pull, and then do a corrector solve.

(c) Move the node below the ground to balance the force.

(d) Nothing; reaction forces do not affect the contact constraints.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 20 **0.5 pts**

Two masses hang in a chain: spring $k_1 = 30\,\mathrm{N/m}$ connects a fixed support to mass $m_1 = 1\,\mathrm{kg}$, and spring $k_2 = 30\,\mathrm{N/m}$ connects m_1 to mass $m_2 = 2\,\mathrm{kg}$. Gravity is $g = 10\,\mathrm{m/s^2}$. At static equilibrium, what is the displacement of m_2 from its position when both springs are unstretched?

(a) $1.00\,\mathrm{m}$

(b) $2.00\,\mathrm{m}$

(c) $1.67\,\mathrm{m}$

(d) $0.67\,\mathrm{m}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 21 **0.5 pts**

A mass m is attached to a spring of stiffness k (no damping, no gravity). Its position $x(t)$ obeys

$$m\ddot{x} + kx = 0.$$

Writing the velocity as $u = \dot{x}$, this is the first-order system $\dot{x} = u$ and $\dot{u} = a$, where $a = -kx/m$ is the acceleration. The *explicit* Euler method advances from time t to $t + \Delta t$ using only the known (old) values at time t:

$$x^{\text{new}} = x^{\text{old}} + \Delta t\, u^{\text{old}}, \qquad u^{\text{new}} = u^{\text{old}} + \Delta t\, a^{\text{old}}, \qquad a^{\text{old}} = -kx^{\text{old}}/m.$$

Let $m = 1\,\text{kg}$, $k = 100\,\text{N/m}$, and $\Delta t = 0.01\,\text{s}$. The mass starts at $x^{\text{old}} = 0.1\,\text{m}$ with $u^{\text{old}} = 0$. After one explicit Euler step, what are x^{new} and u^{new}?

(a) $x = 0.099\,\text{m}$, $u = -0.1\,\text{m/s}$

(b) $x = 0.1\,\text{m}$, $u = 0$

(c) $x = 0.09\,\text{m}$, $u = -1\,\text{m/s}$

(d) $x = 0.1\,\text{m}$, $u = -0.1\,\text{m/s}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 22 **0.5 pts**

An undamped mass–spring system is simulated with explicit Euler (as in Question 21) with any time step Δt. What happens to the oscillation over a long simulation?

(a) Its amplitude grows without bound.

(b) Its amplitude decays to zero.

(c) It matches the exact solution.

(d) The mass stays at its initial position.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 23 **0.5 pts**

An undamped mass–spring system is simulated with implicit Euler, and its oscillation slowly dies out. What happens if the time step Δt is made smaller?

(a) The oscillation dies out faster.

(b) The oscillation dies out more slowly, because the artificial damping is weaker.

(c) Nothing changes; the artificial damping does not depend on Δt.

(d) The simulation becomes unstable.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 24 **0.5 pts**

The Newmark–Beta method advances the position x, velocity $\dot{x}$, and acceleration $\ddot{x}$ from time step n to $n+1$ (step size Δt) using

$$x_{n+1} = x_n + \Delta t\, \dot{x}_n + \Delta t^2 \left[\left(\tfrac{1}{2} - \beta\right) \ddot{x}_n + \beta\, \ddot{x}_{n+1}\right], \qquad \dot{x}_{n+1} = \dot{x}_n + \Delta t \left[(1-\gamma)\, \ddot{x}_n + \gamma\, \ddot{x}_{n+1}\right],$$

where β and γ are parameters that weight the old acceleration $\ddot{x}_n$ against the new one $\ddot{x}_{n+1}$. Which choice of β and γ is unconditionally stable but *strongly dissipative* (the energy of an undamped oscillator decays, similar to implicit Euler)?

(a) $\beta = 0.25$, $\gamma = 0.5$

(b) $\beta = 0$, $\gamma = 0.5$

(c) $\beta = 0.5$, $\gamma = 1.0$

(d) All choices conserve energy.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 25 **0.5 pts**

A metal sphere ($\rho_{\text{metal}} = 7000\,\text{kg/m}^3$) is released from rest in a fluid ($\rho_{\text{fluid}} = 1000\,\text{kg/m}^3$) with no viscosity. With $g = 9.8\,\text{m/s}^2$, what is its speed after 2 s?

(a) 19.6 m/s

(b) 16.8 m/s

(c) 2.8 m/s

(d) 8.4 m/s

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 26 **0.5 pts**

A metal sphere of radius $R = 0.01\,\text{m}$ ($\rho_{\text{metal}} = 10000\,\text{kg/m}^3$) falls in a viscous fluid ($\rho_{\text{fluid}} = 1000\,\text{kg/m}^3$, $\mu = 1000\,\text{Pa}\cdot\text{s}$) with $g = 10\,\text{m/s}^2$. The drag is $6\pi\mu Rv$. What is the terminal speed?

(a) $2 \times 10^{-3}\,\text{m/s}$

(b) $2.2 \times 10^{-3}\,\text{m/s}$

(c) $9 \times 10^{-3}\,\text{m/s}$

(d) $2 \times 10^{-1}\,\text{m/s}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 27 **0.5 pts**

A cantilever of free length $L = 0.1\,\mathrm{m}$, circular cross-section of radius $r_0 = 1\,\mathrm{mm}$, Young's modulus $E = 1\,\mathrm{GPa}$, and density $\rho = 1000\,\mathrm{kg/m^3}$ sags under its own weight ($g = 10\,\mathrm{m/s^2}$). What is the Euler–Bernoulli tip deflection $\delta = qL^4/(8EI)$, with $q = \rho A g$?

(a) $4.0\,\mathrm{mm}$

(b) $0.25\,\mathrm{mm}$

(c) $2.0\,\mathrm{mm}$

(d) $0.5\,\mathrm{mm}$

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 28 **0.5 pts**

In the small-deflection regime, the free length of the cantilever in Question 27 is doubled (all else unchanged). By what factor does the tip deflection change?

(a) 2

(b) 4

(c) 16

(d) 8

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 29 **0.5 pts**

A horizontal cantilever of length L is clamped at one end and sags under its own weight (a uniform load q per unit length). Euler–Bernoulli beam theory predicts the tip deflection $\delta_{\mathrm{EB}} = qL^4/(8EI)$, which grows linearly with q and can even exceed L for a large enough load. A DER simulation with a large load gives a vertical tip deflection much smaller than δ_{EB} (and always less than L). What is the main reason for the difference?

(a) Euler–Bernoulli theory is linear: it assumes small slopes and a shape that stays close to the undeformed one. DER makes no such assumption.

(b) Implicit Euler adds artificial damping that reduces the DER deflection.

(c) The rod stretches a lot at large loads.

(d) DER includes inertia, while Euler–Bernoulli theory does not.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 30 **0.5 pts**

In DER, the bending energy at node k is $\frac{1}{2}EI\,|\boldsymbol{\kappa}_k - \bar{\boldsymbol{\kappa}}_k|^2/\bar{l}_k$, where $\boldsymbol{\kappa}_k$ is the curvature and $\bar{\boldsymbol{\kappa}}_k$ is the natural (stress-free) curvature, and $\bar{l}_k$ is the undeformed Voronoi length of node k. A rod clamped at one end is initialized in the shape of a circular arc, and there is no gravity or other load. Normally $\bar{\boldsymbol{\kappa}}$ is computed from this initial shape, which makes the rod naturally curved. Instead, a student sets $\bar{\boldsymbol{\kappa}}_k = \mathbf{0}$ at every node (a naturally straight rod) but still starts the simulation from the circular arc. What happens?

(a) It keeps its curved shape, because there is no load.

(b) It opens up into a nearly straight rod, because the bending energy is zero only when the rod is straight.

(c) It curls up more tightly.

(d) The simulation fails, because the natural curvature must not be zero.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor

Question 31 **0.5 pts**

Use x, y, z coordinates with gravity along $-z$. The two rods described below are both clamped at one end and simulated with DER. Rod A is naturally straight and initially lies along the x-axis. Rod B is naturally curved: its stress-free shape is a circular arc in the x–y (horizontal) plane, and it starts in that shape. Both rods sag under gravity. Which statement about their twist angles is true?

(a) Both rods twist, because gravity always twists a rod.

(b) Neither rod twists, because the first twist angle is fixed.

(c) Rod A does not twist (it bends only in the x–z plane that contains it); rod B twists (gravity bends it out of the x–y plane in which it is curved).

(d) Rod A twists, but rod B does not.

Space for work (not graded).

Copyright notice: Do not copy, distribute, republish, upload, post or transmit anything from the files hosted on the class website without written permission of the instructor